Tracking the support stiffness of railway wheelset bearings from in-service axle-box acceleration without measuring rail roughness: a multi-pass debiased Whittle likelihood

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ABSTRACT

Loss of the vertical support stiffness of a wheelset bearing, through loose fits, a degraded housing, adapter or pad, or clearance growth, shifts the resonance of the axlebox on its support. On freight wagons this resonance lies in the band of axle-box acceleration excited by rail roughness, so a stiffness identified from it is confounded with the unknown roughness. We pool passes at different speeds and payloads in a single debiased Whittle likelihood: the resonance stays fixed in frequency while the roughness excitation moves with speed, so the roughness levels can be left as free nuisance parameters. A finite-element wheelset is coupled to neighbouring wheels through an infinite Timoshenko rail. For a representative freight wagon (Y25 bogies, ballasted 60E1 track) and 50 passes at 60–110 km/h, simulations give a bias of at most 0.07% and a root-mean-square error of 0.19–0.20%, within 1.08 times the Cramér–Rao bound. Ignoring wheel-to-wheel coupling biases the estimate by -5.6%; nonlinear Hertzian contact shifts it by only 0.059 percentage points. Because absolute values carry a model-form uncertainty of about 10.0%, the method targets changes between periods. Comparing periods within a payload band, with the foundation and contact stiffness re-estimated in each, a decrease of 0.75–0.83% is detectable from 50 empty passes per period and 0.35–0.39% from 200, and a 20% softer foundation causes no bias; pooling mixed payloads requires a payload-dependent suspension model. Clearance growth changes the effective stiffness little (-0.74% per 10 μm), so the method reports support-stiffness loss rather than its mechanism.

1. Introduction

Problem statement. Each journal of a railway wheelset carries the axle load into the axlebox through the rollers of the bearing, the bearing housing, the adapter and, on many bogies, an elastomeric pad. We call the vertical stiffness of this load path the effective support stiffness k_B. It falls when the bearing fits loosen, when the housing, adapter or pad degrades, or when the bearing clearance grows. The problem addressed here is to estimate k_B, and changes of it between monitoring periods, from the vertical axle-box acceleration (ABA) of a wagon in normal service, when the rail roughness that excites the response is not measured.

State of the art. Four families of methods bear on this problem. *Bearing fault diagnostics* detect localised rolling-contact damage from the impacts it produces, using envelope and spectral-kurtosis analysis of high-frequency vibration [1], or from the bearing temperature. They are mature and need no model of the vehicle, but they respond to impacts at roller-passing frequencies, not to a gradual loss of stiffness, so loose fits or a degraded pad remain invisible until secondary damage occurs. *Model-based processing of ABA for the track* uses a vehicle–track model to turn ABA into rail roughness or track stiffness. Roughness is estimated by inverse filtering or source identification with a known vehicle transfer function [2, 3]; track stiffness is estimated from the P2 resonance of single passes [3] or by matching a library of simulations, where the irregularity level enters as one parameter [4]. These methods use the known vehicle as the instrument; they assume the vehicle parameters, and the targets lie below the axle-box support band. Per-pass estimates also scatter widely when the roughness spectrum has features near the resonance used [3]. *Joint identification of vehicle parameters with unknown excitation* has been developed for road vehicles, where the excitation is separated

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from the vehicle by the time lag between axles in a Kalman filter with unknown input [5], by a Bayesian formulation in the frequency domain [6], or by modelling the road as a Gaussian random field [7]. Railway suspension parameters have been identified as a statistical inverse problem with a surrogate model, using measured track irregularities as input [8, 9]. These methods need either the measured excitation or a model of its structure (time lag between axles, random field), and they have been demonstrated for suspension parameters at low frequencies, well below the axle-box support band. *Output-only identification with an unknown excitation spectrum*, such as operational modal analysis with a Bayesian FFT likelihood [10] or drive-by monitoring that removes road roughness by subtracting responses recorded at different speeds [11], treats the excitation as a nuisance in a frequency-domain likelihood. It is statistically efficient, but it assumes a smooth excitation spectrum near each mode and, in its standard form, a single record, so it does not separate a vehicle resonance from a roughness spectrum that is itself unknown and changes from pass to pass.

Motivation. The support resonance of a freight wheelset, the axlebox vibrating on its support, lies at a few hundred hertz, inside the band in which ABA is excited by rail roughness. ABA is already recorded in service for track monitoring, so the support stiffness could be tracked from data that are collected anyway, without a dedicated sensor or test. None of the families above provides this: fault diagnostics do not see stiffness, the track methods assume the vehicle, and the joint-identification methods need the excitation or work below the support band. The work is motivated by condition monitoring of freight wagons on Indian Railways. The vehicle and track values that a model of an Indian wagon needs are not yet available to us and are pending data from the Research Designs and Standards Organisation (RDSO). The method is therefore developed and assessed here for a representative European freight wagon, a four-axle Eaos/Eanos-type wagon on Y25 bogies on ballasted UIC60 track, for which the parameters are published; values that could not be found are swept over plausible ranges.

Principle of the approach. The support resonance is fixed in frequency, while the roughness excitation is fixed in wavenumber and therefore moves in frequency with the train speed. Passes at different speeds thus see the same resonance through different parts of the roughness spectrum. We pool many passes in one frequency-domain likelihood in which the roughness level of each wavenumber sub-band is a free nuisance parameter shared by all passes, payload enters through a weigh-in-motion measurement, and the leakage of the finite records is handled by the debiased Whittle likelihood [12]. The forward model couples the instrumented wheelset to its neighbours through the rail, because the other wheels of the bogie and wagon load the rail at the same time.

Contributions. To our knowledge, this is the first method to identify the vertical stiffness of the axle-box bearing support from in-service axle-box acceleration without measuring rail roughness. The support resonance and the roughness are separated by pooling passes at different speeds and payloads in a single debiased Whittle likelihood, in which roughness levels are free nuisance parameters. Joint estimation of vehicle or track parameters with unknown excitation has been studied for track stiffness [3, 4] and for road-vehicle suspensions [5, 6]. There, roughness is handled by deconvolution, a single level parameter, or axle time lags, and the targets lie below the axle-box support band. Specifically, this paper contributes:

1. a multi-pass debiased Whittle likelihood that identifies the effective support stiffness and damping of a wheelset bearing from ABA spectra of many passes, with the roughness level of each sub-band a free nuisance parameter and payload from a weigh-in-motion measurement;
2. a forward model in which the instrumented wheelset is coupled to its neighbours through an infinite Timoshenko rail with the phase factors between wheels kept, and evidence that ignoring this coupling biases the support stiffness by -5.6%;
3. an error budget for a representative freight wagon: Cramér–Rao bounds and their dependence on speed spread and number of passes, Monte Carlo bias and coverage, model-form sensitivity, sweeps over the published parameter ranges, simulation-calibrated intervals, and a check against nonlinear Hertzian contact;
4. a procedure for detecting changes between monitoring periods within payload bands that re-estimates the track foundation and contact stiffness in each period, with detection limits, false-alarm rates under track drift, and the bearing clearance growth that the detection limit corresponds to.

Importance and impact. A loss of support stiffness changes the load path of the bearing and the dynamic loads on it, and it can develop without the temperature rise or impacts that existing bearing monitoring detects. Detecting it from data that operators already record would add a monitoring channel at no hardware cost. The method needs only

the in-service passes of an instrumented wagon over ordinary track at varying speeds, and its statistical basis gives detection limits and false-alarm rates that can be stated in advance for a given monitoring design. The same likelihood applies to other vehicle–track resonances excited by unmeasured roughness.

Section 2 describes the vehicle–track model and Section 3 the estimator. Section 4 verifies and demonstrates the method in simulation, Section 5 discusses the results and their limitations, and Section 6 concludes.

2. Vehicle–track model

The model represents a four-axle freight wagon with Y25 bogies (22.5 t axle load, 920 mm wheels) on ballasted track with 60E1 rail. Its parameters (Table 1) are taken from published data for such vehicles and track [13–18]; where no published value was found, a range is assumed and the results are swept over it (Section 4.6).

2.1. Wheelset and the effective support stiffness

The wheelset is a stepped Timoshenko beam (axle body, wheel seats, dust-guard collars and journals) discretised with 32 two-node elements with interdependent interpolation, which are free of shear locking [19]. The shear coefficient is Cowper's value for a solid circular section [20]. Each wheel is a rigid disc with mass and diametral inertia at its seat. Each axlebox is a rigid mass on its journal, tied to a bogie frame that is held fixed by a linear primary spring–damper pair. The primary stiffness and damping are the empty-wagon values of the vehicle source [14]. That damping is a low-frequency equivalent viscous value for the friction damper of the Y25 bogie and has not been validated at the 400 Hz of the support resonance, where the damper is likely to stick; the primary damping is therefore treated as uncertain and swept (Sections 4.6 and 5.1). Axle damping is stiffness proportional. The model is linear and vertical-plane. Gyroscopic and rotating-damping terms are omitted: the rotation rate is 0.10 of the first constrained bending frequency at the reference speed (Table 2).

Between each journal and its axlebox the model has one spring–damper pair, k_B and c_B. We call k_B the *effective support stiffness*: the stiffness of the load path from journal to axlebox, that is, the roller contacts of the bearing in series with the bearing housing, the adapter and any pad,

$$k_B(W, c) = \left[\frac{1}{k_{\mathrm{contact}}(W, c)} + \frac{1}{k_{\mathrm{housing}}} \right]^{-1}, \tag{1}$$

where W is the axle load and c the diametral clearance of the bearing. The quantity monitored in this paper is a loss of k_B. It covers loosening of the bearing fits, degradation of the housing, adapter or pad, and growth of the bearing clearance. It is not a model of rolling-contact fatigue or spalling, which act at much higher frequencies through impacts.

Roller contacts. Y25 axleboxes carry two single-row cylindrical roller bearings per journal. Under a static radial load the contact stiffness follows from the line-contact law of the wheelset model, $Q = K_n \delta^{10/9}$, with the load shared over the rollers in the load zone (roller deflection $\delta_j = \Delta \cos\theta_j \cos\alpha - c/2$ with $\alpha = 0$, contact force summed over both bearings; inner and outer contacts in series; 21). For the 130×240×80 mm bearings (18 rollers of 48 mm per row, assumed from the envelope; initial clearance 130 µm, the lower unmounted value; Table 1) the cage-averaged contact stiffness is 3.9 and 6.0 GN/m per journal at the empty and loaded axle loads. These values are several times the nominal effective support stiffness of 1 GN/m used throughout, so in reading (1) the housing path carries most of the compliance. A change of the contact stiffness then reaches k_B diluted by the factor k_B/k_{contact}, 0.26 (empty) to 0.17 (loaded): a clearance growth of 10 µm lowers the contact stiffness by -2.8% (empty) and -1.5% (loaded), and k_B by -0.74% and -0.25%.

If instead k_B were the contact stiffness alone, the axlebox-on-bearing resonance would sit near 1229 Hz (single-mass estimate with the 100 kg axlebox), where contact filtering, the rail pinned–pinned mode and the flexible modes of the wheels, none of which the model contains, dominate; that reading is outside the scope of this paper.

Load dependence. The default stiffness law used for all results is the power law

$$k_B(W) = k_{B0}\,(W/W_0)^p, \qquad p = 0.1, \tag{2}$$

the exponent of a line contact at zero clearance, with W_0 the loaded axle load and k_{B0} the health parameter. The series law (1), with k_{housing} set so that $k_B(W_0, c_0) = k_{B0}$, is implemented as an option; its effective exponent over the payload range is 0.061. The exponent is held fixed in the estimator; Section 5.1 discusses why it does not serve as a second, clearance-specific indicator.

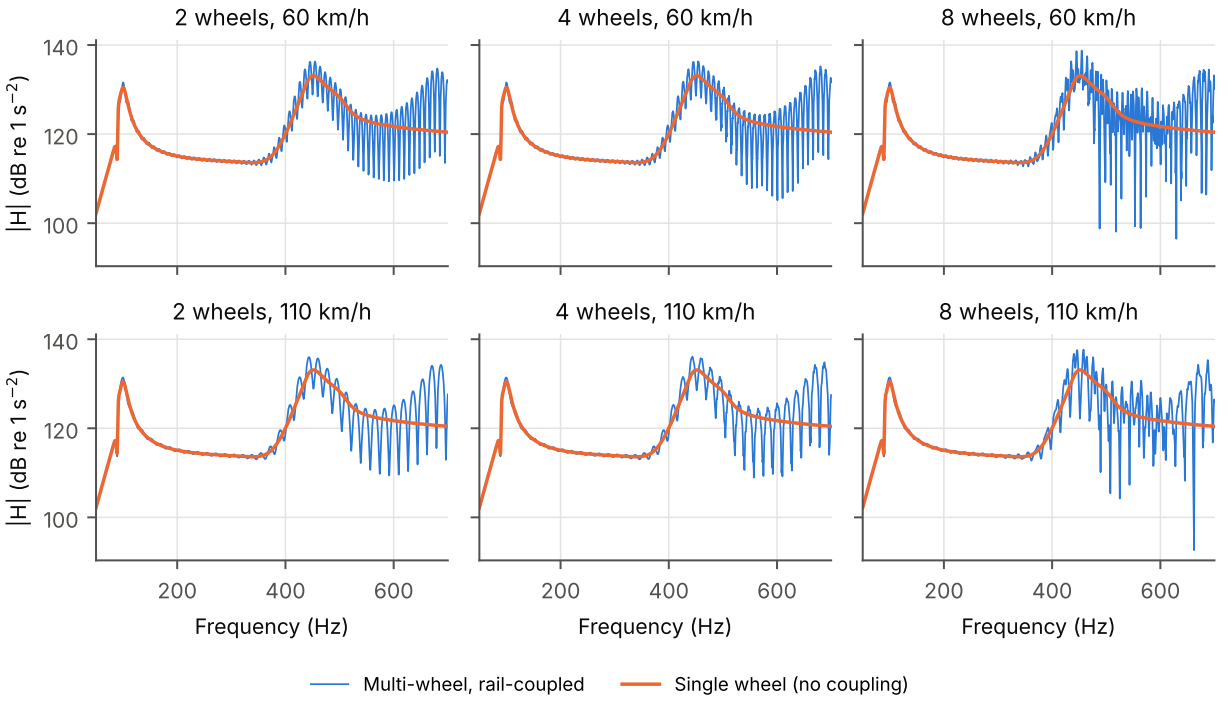


Figure 1: Transfer functions with and without rail coupling. Axle-box acceleration per unit roughness, $|H|$, for 2, 4 and 8 rail-coupled wheels against a single wheel without coupling, at 60 and 110 km/h. Data: `results/p2/fig1_transfer.json`.

2.2. Rail, contact and multi-wheel coupling

The rail is an infinite Timoshenko beam on a continuous foundation of complex stiffness per unit length $k_f(1+\mathrm{i}\eta_f)$. Its point and transfer receptances $R(x)$ follow in closed form by partial fractions of the wavenumber-domain solution; the result equals Eq. (11) of Carrigan and Talbot [3] and is checked against a numerical inverse transform. The representative track is ballasted with 60E1 rail; the foundation stiffness per unit length is the rail pad stiffness divided by the sleeper spacing [18], and its range, the loss factor and the contact stiffness are given in Table 1. Each wheel meets the rail through a linearised Hertzian spring $k_H = k_{H0}(W/W_0)^{1/3}$ [3]. The reference value k_{H0} follows from Hertz theory for elliptical contact [22] between an S1002 wheel of rolling radius 0.46 m and a UIC60 rail of crown radius 0.30 m at the 110 kN static wheel load, with the transverse radius of the wheel tread taken as infinite, giving $k_{H0} = 1.49$ GN/m; tread radii between +0.5 and −1.0 m give 1.39–1.58 GN/m, within the range reported in the literature [23].

The instrumented wheelset runs with neighbouring wheels at positions d_k along the same rail. With $\mathbf{A}^w$ the wheel receptances, $\mathbf{A}^c = \mathrm{diag}(1/k_H)$ and $A^r_{jk} = R(|d_j - d_k|)$, the contact forces under a roughness $z(x)$ are

$$\mathbf{F} = (\mathbf{A}^c + \mathbf{A}^r + \mathbf{A}^w)^{-1}\mathbf{Z}, \qquad Z_k = Z\exp(\mathrm{i}\omega d_k/V), \tag{3}$$

[3, Eqs. 5 and 8], and the axle-box acceleration per unit roughness at the instrumented wheel is

$$H(f;\theta,W,V) = \sum_k G_k(f;\theta,W)\exp(\mathrm{i}2\pi f d_k/V). \tag{4}$$

Carrigan and Talbot drop the phase factors (their Eq. 6) because their spectra cannot resolve them. We keep them: with records of 20 wheel revolutions the frequency resolution resolves the ripples of period $V/(d_k - d_j)$ (Fig. 1), and Section 3.2 handles their effect on the likelihood. The default configuration has four wheels: the instrumented bogie and the next bogie of the same wagon, at the Y25 wheelbase and the bogie pivot pitch of a four-axle Eanos wagon (Table 1). The layout is a configuration parameter; up to eight wheels (two coupled wagons) are used in Section 4.3.

Table 1
Representative freight wagon and track: model parameters and sources

Group	Parameter	Value	Range	Unit	Source
Material	E	2.1e+11	–	Pa	doc §16 (standard steel)
Material	nu	0.3	–	-	doc §16
Material	rho	7850	–	kg/m^3	doc §16
Axle	length	2.18	2.18–2.188	m	[snippet] BA004 freight wheelset (Y25), total axle length 2180 mm (railwayspare-parts.com listing); 2187.6 mm Type D a
Axle	sections [outer ξ, d]	[[0.64,0.173],[0.86,0.2],[0.9,0.16],[1.09,0.13]]	–	m	diameters [snippet]: body 173 mm, wheel seat 200 mm, journal 130 mm (BA004 listings; Unirail BA004, EN 13261 130x1
Axle	wheel_xi	0.75	–	m	[snippet] 2s = 1500 mm between nominal rolling circles (Bizic & Petrovic 2021, Mechanics Transport Communications 19
Axle	journal_xi	1	–	m	[snippet] 2b = 2000 mm between axle-box centres (Bizic & Petrovic 2021)
Axle	n_elements	32	–	-	brief: 20-40 elements; convergence checked at 20/30/40
Axle	damping_zeta	0.01	0.001–0.02	-	[snippet] measured damping ratios of the first torsion and first and second bending modes below 2% (Chaar 2007, KTH d
Axle	damping_fref	96	96–115	Hz	[snippet] first wheelset bending mode of a freight wagon 96 Hz (Sci. Rep. 2025, s41598-025-15781-4); 115 Hz (Mu et al.
Wheel	mass	304	–	kg	[snippet] 920 mm ER7 wheel, nominal weight 304 kg (Unirail BA004 and BA 092 wheelset pages)
Wheel	diametral_inertia	20	10–30	kg m^2	assumed, swept (single-wheel inertia not found; disc estimate m r^2/4)
Wheel	r0	0.46	0.42–0.46	m	full text verified: rolling radius 0.46 m, S1002 wheels on UIC60 rail (Dusza, Golofit-Stawinska & Zboinski 2024, Appl. Sci.
Support	kB0	1e+09	5e+08–5e+09	N/m	assumed, swept (effective support stiffness of a freight axlebox not found; indications [snippet]: equivalent vertical stiffnes
Support	cB	6740	3000–1.2e+04	N s/m	assumed, swept (round-1 snippet attributed 6740 N s/m to Li et al. 2019, Adv. Mech. Eng. 11(4), 10.1177/16878140198
Support	load_exponent	0.1	–	-	doc Eq. (12): Q ~ delta^(10/9) gives k ~ Q^(1/10)
Bearing (health check)	rollers_per_row	18	15–21	-	assumed from the envelope (pitch circle pi x 185 mm / ~1.15 x 27.5 mm roller pitch), swept
Bearing (health check)	roller_length	48	40–56	mm	assumed, ~0.6 x 80 mm bearing width, swept
Bearing (health check)	roller_diameter	27.5	–	mm	assumed, (240 - 130)/4 mm; does not enter the line-contact law
Bearing (health check)	pitch_diameter	185	–	mm	envelope: (130 + 240)/2 mm [snippet: WJ 130x240x80 listings]
Bearing (health check)	contact_angle	0	–	deg	cylindrical roller bearing
Bearing (health check)	clearance_d0	130	0–180	um	[snippet] radial clearance of WJ/WJP 130x240 P before mounting 130-180 um (SKF Railway Technical Handbook; Schae
Axlebox	mass	100	100–158.4	kg	full text verified: axlebox 100 kg (Dusza et al. 2024, Table A1); 158.42 kg [snippet] (Bustos et al. 2025, Y21 bogie rig mo
Primary	kp	1.017e+06	1.017e+06–2.28e+06	N/m	full text verified: Y25 primary vertical stiffness 1017 kN/m empty / 2280 kN/m loaded per axlebox (Dusza et al. 2024, Ta
Primary	cp	7000	7000–1.233e+05	N s/m	full text verified: Y25 primary equivalent vertical damping 7 kN s/m empty / 123.3 kN s/m laden (Dusza et al. 2024, Tab
Payload	W_min	5.5	5.8–6.6	t	design value 5.5 t; [snippet] Eanos tare 23.1 t (Wascosa), 23.5 t (DB Cargo), 24.0 t (Greenbrier), 24.5-26.5 t (VTG, Tatra
Payload	W_max	22.5	–	t	[snippet] Y25 bogie axle load 22.5 t (railway-technology.com, Y25 bogie)
Payload	W0	22.5	–	t	reference load for kB0 (loaded)
Rail 60E1	A	0.00767	–	m^2	[snippet] 76.70 cm^2, mass 60.21 kg/m (ArcelorMittal 60E1 datasheet, EN 13674-1; Valente; British Steel rail profiles da
Rail 60E1	I	3.038e-05	–	m^4	[snippet] I_xx 3038.3 cm^4 (British Steel and ArcelorMittal datasheets, EN 13674-1)
Rail 60E1	kappa	0.4	0.34–0.4	-	[snippet] rail shear coefficient 0.4 (TU Delft thesis, rail close to UIC54); 0.393 for UIC60 (attribution weak); 0.34 BS113A
Track (ballasted)	pad_stiffness	2e+08	8e+07–8e+08	N/m	[snippet] typical UIC60 ballasted track, rail pad 200 MN/m per pad (Thompson, Jones, Waters & Farrington 2007, Appl.
Track (ballasted)	sleeper_spacing	0.6	0.6–0.65	m	[snippet] 0.6 m (EUROBOLT optimal track); 0.65 m (CT23 Table 1; Thompson et al.)
Track (ballasted)	kf	3.333e+08	1.23e+08–1.333e+09	N/m^2	derived: pad stiffness / sleeper spacing (pad modulus, PWI Guide to Track Stiffness definition) from the sourced ranges ab
Track (ballasted)	eta_f	0.2	0.1–0.25	-	[snippet] rail pad loss factor 0.16 (Thompson et al. 2007 table), 0.25 (rolling-noise models), about 0.1 (EVA) to 0.2 (natu
Contact	kH0	1.489e+09	1.388e+09–2.3e+09	N/m	Hertz theory at W0 = 22.5 t (F0 = 110 kN), results/p4/contact_hertz.json; 1.388-1.576e9 for S1002 tread transverse rad
Contact	C_H	9.413e+10	–	N/m^1.5	Hertz theory (Johnson 1985): S1002 wheel, rolling radius 0.46 m (Dusza et al. 2024, full text verified), on UIC60 rail, crow
Layout	wheel positions	[0.0,1.8,10.7,12.5,15.74,17.54,26.44,28.24]	–	m	derived from [snippet] Y25 wheelbase 1800 mm (railway-technology.com; Tatravagonka Y25 data) and Eanos bogie pivot

8 of 38 entries were not found in the literature (marked 'assumed, swept'); their effect is in Table 15. [snippet] = value seen in search results; the full-text status of each source is in docs/parameter_sources.md.

Table 2
Verification of the reference model

Check	Measured	Threshold	Gate
Free–free Timoshenko frequencies vs closed form (max rel. error, 4 modes)	3.62e-03	5e-03	PASS
Energy balance, windowed integral (max rel. residual, 3 cases)	6.7e-11	1e-06	PASS
Frequency vs time domain, axlebox accel. (max rel. L2, 3 cases)	1.8e-06	1e-03	PASS
Mesh 20 vs 40 elements, modes (max rel. diff.)	2.0e-03	–	–
Gyroscopic neglect, Ω/ω_1	0.101	–	–

Rail receptance vs CT23 Eq. 11 and the numerical inverse transform, and the single-wheel coupling identity (1e-9), are unit tests in tests/test_track.py (all pass).

2.3. Verification

Table 2 lists the checks of the reference model: free–free Timoshenko frequencies against the closed form, the windowed energy balance of the steady-state solution, frequency- against time-domain response, and mesh convergence. Unit tests cover the rail receptance against Eq. (11) of [3] and the numerical inverse transform, and the identity of the single-wheel coupled model with the rail-oscillator model.

3. Estimation method

3.1. Observation model

Pass i is a record of vertical axle-box acceleration over 20 wheel revolutions, so its length is fixed in space, $L = 20 \cdot 2\pi r_0$, and its duration $T_i = L/V_i$ depends on the pass speed V_i. The DFT coefficient c_{ik} at bin k belongs to the frequency $f_{ik} = k/T_i$ and to the same wavenumber $\gamma_k = 2\pi k/L$ on every pass. The analysis band is fixed in frequency.

Rail roughness is a stationary Gaussian process with one-sided wavenumber spectrum

$$G(\gamma) = G_{\mathrm{ISO}}(\gamma)\, g_{s(\gamma)}, \tag{5}$$

where G_{ISO} has the shape of the ISO 3095 limit spectrum [24] and g_s is a free level in sub-band s, with K sub-bands per one-third octave. The levels are nuisance parameters: the method never measures roughness. The axle load of each pass, W_i, is a nuisance with a Gaussian prior on $\log W_i$ centred on a weigh-in-motion reading with standard deviation 3%. The sensor noise is white with level e_i, estimated from an out-of-band part of the same record. The globals $\boldsymbol{\theta}$ are the health parameters (k_{B0}, c_B), optionally with track parameters such as k_f and k_{H0}.

Under this model the coefficients are approximately independent complex Gaussians with variance

$$v_{ik} = |H(f_{ik}; \boldsymbol{\theta}, W_i, V_i)|^2\, 2G(\gamma_k)\, \Delta\gamma + e_i, \tag{6}$$

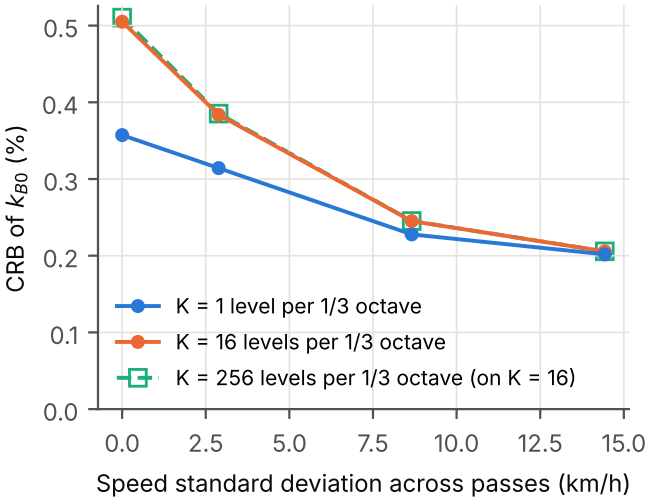


Figure 2: Effect of speed spread on identifiability. Cramér–Rao bound of k_{B0} against the speed spread across passes, for K roughness levels per one-third octave ($N = 50$, 40 dB, payload spread). Data: `results/p1/bound_speed_subband.json`.

and the negative log-likelihood over N passes is the multi-pass Whittle form [25]

$$\ell(\boldsymbol{\theta}, \boldsymbol{g}, \boldsymbol{W}) = \sum_{i=1}^{N} \sum_{k} \left[\log v_{ik} + \frac{|c_{ik}|^2}{v_{ik}} \right] + \sum_{i=1}^{N} \frac{(\log W_i - \log \hat{W}_i)^2}{2\sigma_W^2}. \tag{7}$$

The same structure, with the modal-force spectrum per band as the nuisance, underlies Bayesian FFT operational modal analysis [10].

Why speed diversity identifies the support. Roughness is fixed in wavenumber, while the resonance of the axlebox on its support is fixed in frequency. A pass at speed V_i maps wavenumber γ to frequency $V_i\gamma/2\pi$, so passes at different speeds see the same roughness sub-band through different parts of $|H|^2$. With a single speed, a change of $|H|^2$ and a change of roughness level are indistinguishable sub-band by sub-band; with a spread of speeds they separate. Figure 2 shows the Cramér–Rao bound of k_{B0} falling with the speed spread for several sub-band resolutions K.

3.2. Debiased likelihood for multi-wheel cross terms

Equation (6) samples $|H|^2 G$ at the bin. The expected periodogram of a finite record is instead the spectrum convolved with the Fejér kernel $T\,\mathrm{sinc}^2((f - f_k)T)$. The two agree when the spectrum is smooth on the scale of one bin, as for a single wheelset. They do not agree once the multi-wheel phase factors in (4) put ripples of period $V/(d_k - d_j)$ close to the bin width. Following Sykulski et al. [12], we keep the Whittle form but replace v_{ik} by the exact expected periodogram,

$$v_{ik} = \sum_{m} \frac{\mathrm{sinc}^2\big((f_m - f_{ik})T_i\big)}{M} \, |H(f_m)|^2 \, 2G(\gamma_m)\,\Delta\gamma + e_i, \tag{8}$$

with the sum over a grid $M = 4$ times finer than the bins, truncated at ± 120 bins (a narrower truncation at ± 30 bins leaves a measurable bias; Section 4.4). On the simulated 12-wheel setup of [3], the bin-sampled model differs from the mean of 300 periodograms by an rms log-ratio of 0.47; the debiased model differs by 0.07, against 0.06 expected from sampling alone (Fig. 3).

3.3. Profiling and optimisation

For given globals $\boldsymbol{\theta}$, the nuisances ($\log \boldsymbol{g}$, $\log \boldsymbol{W}$) are profiled out by Fisher scoring, warm-started from the previous evaluation. The globals are then found by Nelder–Mead on the profile likelihood. With the debiased variance (8) the

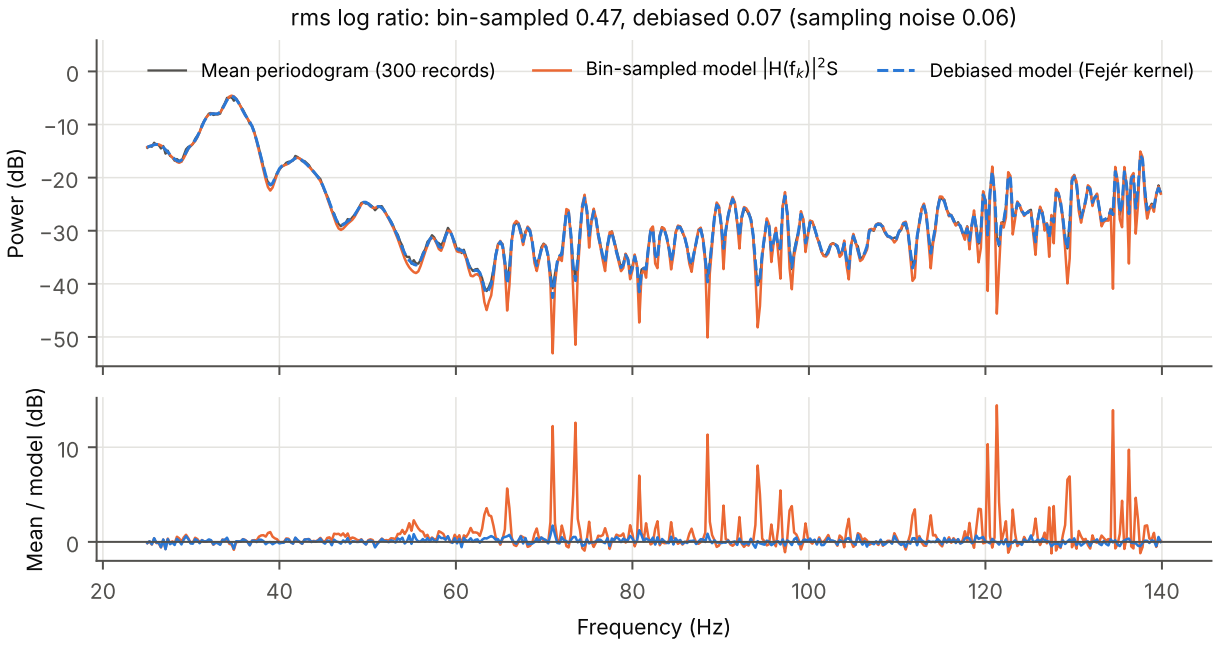


Figure 3: Check of the debiased likelihood. Mean periodogram of 300 records against the bin-sampled and the debiased (Fejér-convolved) models (top) and their ratio (bottom). Data: `results/p2/debias_check.json`.

kernel mixes neighbouring roughness sub-bands. The default scoring step attributes each bin's variance to its own sub-band; the exact step uses the full derivative $\partial v_{ik}/\partial \log g_s$ over all sub-bands and the corresponding Fisher matrix, so that its fixed point is the exact profile maximum. Section 4.4 compares the two.

The transfer functions are computed directly from the finite-element and rail models (no surrogate): a table of the coupled components $G_k(f;\boldsymbol{\theta},W)$ over payload nodes, built from an exact modal decomposition of the wheelset for each trial $\boldsymbol{\theta}$, is interpolated in $\log W$ and frequency.

3.4. Uncertainty

Wald intervals use the observed information of the profile likelihood (numerical Hessian). Profile-likelihood intervals use the χ^2_1 cut, re-optimising the other globals and all nuisances. The Cramér–Rao bound (CRB) is the inverse expected Fisher information with every nuisance free and the weigh-in-motion prior included. When records repeat over the same section, the independence assumed in (7) fails across passes; the variance is then taken from a sandwich estimator clustered on DFT bin (the same wavenumber on every pass). Where the Wald intervals under-cover in simulation (Section 4.1), they are simulation-calibrated for the design: the mean bias of $\log \hat{k}_{B0}$ found by simulation is subtracted and the standard error is multiplied by the simulated ratio of RMSE to standard error, checked on held-out simulation seeds.

3.5. Relative change and detection

The monitoring output is the change of support stiffness between a baseline period and a later period of the same vehicle and sensor, each with N passes in the same payload band. Two bands are used: empty (axle load 5.5–7 t) and laden (20–22.5 t); in each, the estimator uses the band's primary stiffness and damping (Table 1), and the weigh-in-motion payload assigns each pass to its band. Pooling passes over the whole payload range requires a payload-dependent model of the primary suspension (Section 4.8). The change is

$$D = \log \hat{k}_{B0}^{\,\text{later}} - \log \hat{k}_{B0}^{\,\text{base}}, \tag{9}$$

reported as $\Delta k_{B0}/k_{B0} = \exp(D) - 1$. A misspecified model parameter biases $\log \hat{k}_{B0}$ in both periods; to first order the common part cancels, and the remainder scales with the change itself. We therefore use the interval

$$D \pm 1.645\sqrt{\mathrm{se}_{\mathrm{base}}^2 + \mathrm{se}_{\mathrm{later}}^2 + (\kappa D)^2}, \tag{10}$$

where κ is the model-form gain uncertainty: the root sum of squares, over the uncertain model parameters, of the gain $\partial(\hat{D}/D)/\partial \log(\text{parameter})$ times that parameter's 1σ log range. Foundation stiffness can drift between periods (seasonal pad stiffness), so the default fit re-estimates k_f and k_{H0} in each period together with (k_{B0}, c_B).

A loss is declared when $D < -1.645\, s_0$, where s_0 is the standard deviation of D with no change (5% false-alarm rate, one-sided). The minimum detectable change (MDC) is the true decrease detected with probability 0.9, MDC $= (1.645 + 1.282)\, s_0$ without bias; with a bias in D it is found by interpolating the bias against the true change.

3.6. Model-form bias

The large-sample bias under a misspecified model is the pseudo-true parameter: the minimiser of the expected value of (7) when the periodograms are replaced by their expectations under the true model. It is computed by the same profiled fit with $|c_{ik}|^2$ replaced by the true v_{ik}, and was checked against full Monte Carlo fits. A first-order (score) approximation was found to fail for large parameter errors and is not used.

3.7. Implementation and use of AI

The simulation and estimation code was developed with the assistance of an AI coding tool, Claude Code (developer: Anthropic; model: Claude Opus 5.5). It was used to write the code (the finite-element and rail models, the simulation and estimation scripts, and the scripts that generate tables, figures and the numbers quoted in the text) and to assist the literature search. All code was reviewed, tested and verified by the authors: unit tests cover the finite-element and rail models, the model is verified against closed-form solutions (Table 2), and every result in this paper is regenerated by script from stored, version-controlled result files.

4. Results

The results are organised in two parts. *Verification* establishes that the estimator does what it claims under controlled conditions: the reference model against closed-form solutions (Table 2); the debiased likelihood against simulated periodograms (Fig. 3); identifiability through the Cramér–Rao bound and its dependence on speed spread (Section 4.2); bias, scatter and interval coverage in a five-seed Monte Carlo study (Section 4.1) with a diagnosis of the residual bias (Section 4.4); and robustness to model errors, to the parameter ranges of the representative vehicle and to nonlinear contact (Sections 4.5, 4.6 and 4.7). *Demonstration* applies the method to the monitoring task: detecting a change of support stiffness between periods (Section 4.9), what that detection limit means for bearing clearance growth (Section 4.10), false alarms under track drift (Section 4.11), and practical variants of the measurement (Sections 4.12–4.14).

All results are synthetic: data are generated by the reference model of Section 2 with the roughness, speed, payload and noise model of Section 3.1, and fitted with the estimator of Section 3. The default design is $N = 50$ passes with speeds uniform on 60–110 km/h, payloads uniform on 5.5–22.5 t with a 3% weigh-in-motion error, four rail-coupled wheels, and $K = 16$ roughness levels per one-third octave drawn with a 3 dB spread. The frozen estimator uses the debiased likelihood with a $\pm$120-bin kernel, own-sub-band scoring, $K = 16$ and a fixed load exponent. Unless stated otherwise, results are for the representative wagon of Section 2 with the values of Table 1. The headline Monte Carlo study was started before the contact stiffness was computed from Hertz theory and the primary damping taken from the vehicle source, and uses $k_{H0} = 1.4$ GN/m and $c_p = 10$ kN s/m; both lie within the swept ranges of Table 7, where they change the bound by a few hundredths of a percentage point. Three studies were run with an earlier configuration (a metro wheel layout, BS113A rail and a $\pm$30-bin kernel) and are reported for their qualitative conclusions only: the bias diagnosis (Section 4.4), the multi-wheel Monte Carlo (Fig. 4), the gain uncertainty of relative changes and its replicates (Section 4.9). Sections 4.2, 4.12 and 4.13 use a single wheelset on a rail oscillator, whose transfer function is smooth on the bin scale, and Section 4.14 uses the setup of [3].

4.1. Headline Monte Carlo

Table 3 summarises 50 datasets per noise level (five seeds of ten). At 20 dB the estimate of k_{B0} has a bias of -0.01 $\pm$ 0.03% and an RMSE of 0.20%, 1.04 times the Cramér–Rao bound; Wald 90% intervals cover the truth in 0.88

Table 3
Headline Monte Carlo, 4-wheel rail-coupled model, frozen estimator (k_B0)

SNR (dB)	n	Bias (%) ± SE	RMSE (%)	CRB (%)	RMSE/CRB	RMSE/SE	Wald 90% coverage [95% CI]	Simulation calibration: bias b, inflation r; held-out coverage	Ignore coupling (seed 42, 10 datasets): bias (%) ±
20	50	-0.01 ± 0.03	0.20	0.20	1.04	1.04	0.88 [0.76, 0.95]	none (uncalibrated coverage ≥ 0.85)	–
40	50	0.07 ± 0.02	0.19	0.17	1.08	1.07	0.84 [0.71, 0.93]	b = +0.069%, r = 1.07; split A 0.80 [0.56, 0.94], split B 0.97 [0.83, 1.00]	-5.55 ± 0.20 (covered 0/10)

N = 50 passes, 60–110 km/h, 5 seeds × 10 datasets. Frozen estimator: debiased Whittle, kernel ±120 bins, own-sub-band scoring, K = 16, load exponent fixed. Split A calibrates on seeds 42/123/456 and tests on 789/1024; split B the reverse.

Table 4
Effect of speed spread and number of passes (single wheelset, k_B0)

N	SNR (dB)	Speeds (km/h)	Datasets	Bias (%) ± SE	RMSE (%)	CRB (%)	RMSE/CRB	Wald 90% cov. [95% CI]
50	20	fixed 100	20	-0.10 ± 0.18	0.80	0.66	1.22	0.80 [0.56, 0.94]
50	20	95–105	20	-0.06 ± 0.11	0.47	0.44	1.05	0.95 [0.75, 1.00]
50	20	60–110	20	0.09 ± 0.06	0.26	0.23	1.12	0.85 [0.62, 0.97]
50	40	fixed 100	20	-0.09 ± 0.15	0.64	0.66	0.97	0.90 [0.68, 0.99]
50	40	95–105	20	0.07 ± 0.07	0.33	0.47	0.70	1.00 [0.83, 1.00]
50	40	60–110	20	0.09 ± 0.04	0.20	0.23	0.89	0.95 [0.75, 1.00]
200	40	fixed 100	20	0.10 ± 0.07	0.31	0.32	0.96	0.95 [0.75, 1.00]
200	40	95–105	20	0.04 ± 0.05	0.22	0.22	0.98	0.95 [0.75, 1.00]
200	40	60–110	20	0.00 ± 0.03	0.11	0.11	0.95	0.90 [0.68, 0.99]

of datasets [0.76, 0.95]. At 40 dB the bias is 0.07 ± 0.02%, the RMSE 0.19% (1.08 times the bound) and the Wald coverage 0.84 [0.71, 0.93]. The damping c_B is less well determined and biased by 5.0% and 7.1%. Section 4.4 reports what was done to explain the residual bias and excess scatter.

Simulation-calibrated intervals. Because the 40 dB intervals under-cover, they are calibrated by simulation for this design: the mean bias of $\log \hat{k}_{B0}$ is subtracted (0.07%) and the Wald standard error is multiplied by the measured ratio of RMSE to standard error (1.07). At 20 dB the uncalibrated coverage is 0.88 and no calibration is applied. Calibrating on three seeds and testing on the other two, the held-out coverage is 0.80 (16/20; [0.56, 0.94]) with the inflation estimated as 0.99, and 0.97 (29/30; [0.83, 1.00]) for the reverse split (inflation 1.18). At 20 dB the held-out coverage without calibration is 1.00 (20/20) and 0.70 (21/30; [0.51, 0.85]) for the two splits: the pooled coverage meets the threshold, but it varies between groups of seeds. These intervals are valid for the simulated design only.

Ignoring the coupling. Fitting data of the four-wheel coupled model (seed 42, 40 dB, 10 datasets) with a single wheelset on the same rail, i.e. without the neighbouring wheels, biases k_{B0} by -5.6 ± 0.2%, and 0 of 10 Wald intervals cover the truth.

4.2. Speed spread and number of passes

The bound on k_{B0} is set by the speed spread (Table 4, Fig. 2). For $N = 50$ at 40 dB it is 0.66% with all passes at 100 km/h, 0.47% for speeds within 95–105 km/h and 0.23% for 60–110 km/h; with $N = 200$ and the wide spread it falls to 0.11%.

4.3. Multi-wheel coupling

With 60–110 km/h, the bound on k_{B0} is 0.161–0.166% for one to eight coupled wheels of the representative wagon (Table 5); the coupling neither helps nor hurts identifiability (0.77–0.91% at a single speed). It does matter for bias (Fig. 4). In a Monte Carlo study with the earlier configuration (a metro wheel layout, BS113A rail and a ±30-bin kernel), fitting multi-wheel data with a single-wheelset model biased k_{B0} by -13.3%, -13.9% and -8.6% for 2, 4 and 8 wheels, while the coupled model gave 0.24 ± 0.07%, 0.12 ± 0.04% and 0.04 ± 0.04%. With the Y25/Eanos layout of the representative wagon, the four-wheel bias of the single-wheelset model is -5.6% (Section 4.1): smaller, but still far outside the statistical error. The size of the coupling bias therefore depends on the wheel layout, and the coupled model is needed in either case.

4.4. Estimator bias diagnosis

The diagnosis uses the ten seed-123 datasets at 40 dB, where the bias is largest (0.277% with the ±30-bin kernel, RMSE/bound 1.83), and seed 42 as a contrast (0.021%, 1.33); Table 15.

- *Noise.* The same datasets at 60 dB give 0.284% (RMSE/bound 1.87); per dataset the estimates change by at most 0.02 percentage points. The error is fixed by the roughness, speed and payload realisation, not by sensor noise.
- *Scoring and table resolution.* Exact sub-band scoring with a 0.25 Hz table gives 0.274% (at most 0.01 percentage points from the default).

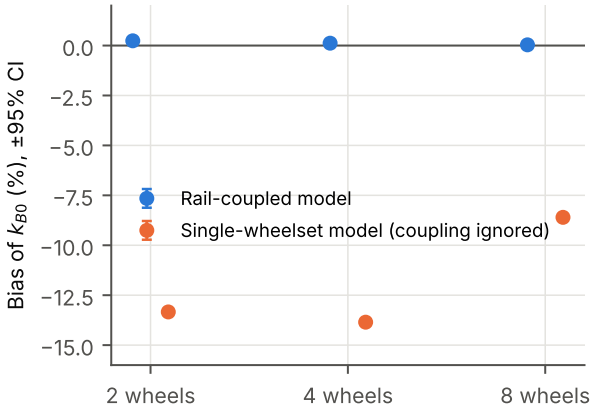


Figure 4: Effect of ignoring wheel-to-wheel coupling. Bias of k_{B0} with the rail-coupled model and with a single-wheelset model, for 2, 4 and 8 coupled wheels (95% intervals), earlier configuration (metro wheel layout). Data: `results/p2/multiwheel_mc.json`.

Table 5
Multi-wheel coupling through the rail (k_B0)

(a) Bound

Rail	Wheels	Speeds	N	SNR (dB)	CRB k_B0 (%)	CRB c_B (%)
60E1	1	fixed_100	50	40	0.811	36.9
60E1	1	uniform_95_105	50	40	0.398	18.6
60E1	1	uniform_60_110	50	40	0.166	7.5
60E1	2	fixed_100	50	40	0.822	37.3
60E1	2	uniform_95_105	50	40	0.411	19.2
60E1	2	uniform_60_110	50	40	0.161	7.3
60E1	4	fixed_100	50	40	0.906	41.2
60E1	4	uniform_95_105	50	40	0.401	18.8
60E1	4	uniform_60_110	50	40	0.164	7.4
60E1	8	fixed_100	50	40	0.774	35.0
60E1	8	uniform_95_105	50	40	0.432	20.2
60E1	8	uniform_60_110	50	40	0.165	7.5

(b) Monte Carlo, 60–110 km/h, N = 50, 40 dB

Wheels	Fitted model	Datasets	Bias (%) ± SE	RMSE (%)	RMSE/CRB	Wald 90% cov. [95% CI]
2	rail-coupled	10	0.24 ± 0.07	0.32	1.91	0.50 [0.19, 0.81]
2	single wheelset	10	-13.34 ± 0.08	14.32	85.89	0.10 [0.00, 0.45]
4	rail-coupled	10	0.12 ± 0.04	0.17	0.98	0.90 [0.55, 1.00]
4	single wheelset	10	-13.85 ± 0.07	14.91	87.88	0.00 [0.00, 0.31]
8	rail-coupled	10	0.04 ± 0.04	0.14	0.81	0.90 [0.55, 1.00]
8	single wheelset	10	-8.60 ± 0.09	9.00	52.86	0.00 [0.00, 0.31]

The 2-wheel rail-coupled bias is an open item (docs/paper1_outline.md).

- *Kernel truncation.* Widening the debiasing kernel from ±30 to ±120 bins lowers every estimate, to 0.219% (RMSE/bound 1.53): about a fifth of the bias. The wide kernel is the default.
- *Number of roughness levels.* With the wide kernel, $K = 4$ gives 0.239% (RMSE/bound 1.76); $K = 1$ cannot represent the 3 dB sub-band variation of the data and fails (RMSE/bound 5.22). Too many profiled nuisance levels is not the cause.
- *Large-sample bias.* With the correct model, pseudo-true fits reproduce relative changes of −2, −5 and −10% with slope 0.99998.
- *Correlation between bins.* A sandwich standard error clustered on 61-bin blocks widens the 40 dB standard error from 0.170% to 0.179% and raises the coverage only from 0.68 to 0.74 (fits with the ±30-bin kernel).
- *Pass design.* Across the 50 headline datasets, speed spread, payload spread, minimum speed and the number of bins near the support resonance explain $R^2 = 0.19$ of the error (joint $p = 0.049$; payload spread alone $p = 0.042$, not significant after four tests).

4.5. Model-form sensitivity

Figure 5 and Table 6 give the large-sample bias of k_{B0} when one model parameter is wrong. The unsprung mass dominates (-16.6% per +10% mass error), followed by the foundation stiffness (2.00% per +10%). At realistic 1σ ranges the absolute budget is 21.3% (root sum of squares), or 20.1% with the unsprung mass known to ±1%; re-estimating k_f

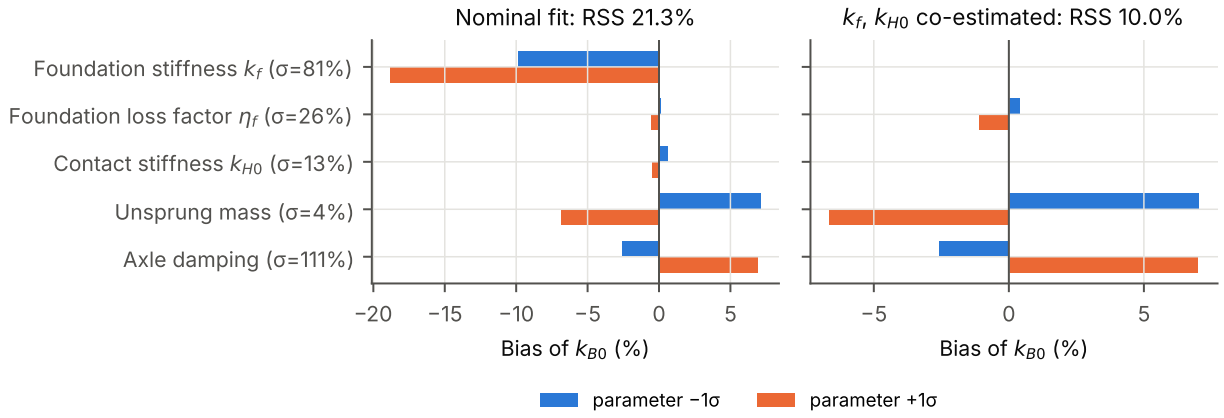


Figure 5: Model-form sensitivity. Bias of k_{B0} at $\pm 1\sigma$ errors of each uncertain model parameter, with k_f and k_{H0} fixed (left) and re-estimated (right). Data: `results/p2/model_form_mw_sensitivity.json`.

Table 6
Model-form sensitivity and absolute bias budget (rail-coupled model)

Parameter	1σ (log)	Range source	Bias per +10% (%)	Nominal: bias at −1σ / +1σ (%)	k_f, k_H0 co-estimated: bias at
kf	0.60	rail pad 80-800 MN/m per pad at 0.60-0.65 m spacing -> 123-1333 MN/m^2 (Thompson et al. 2007; EUROBOLT; snippets)	+2.00	-9.87 / -18.81	+0.00 / -0.00
eta_f	0.23	rail pad loss factor 0.1-0.25 (snippets)	-0.14	+0.12 / -0.55	+0.40 / -1.08
kH0	0.13	linearised contact stiffness 1.388 GN/m (Hertz, S1002 transverse radius +0.5 m) to 2.3 GN/m (published spread, Hasan 2020)	-0.44	+0.62 / -0.46	-0.00 / +0.00
unsprung_scale	0.04	about 20 kg per axlebox, 4% of the wheelset unsprung mass, between axlebox designs (SKF Evolution, snippet)	-16.55	+7.10 / -6.81	+7.04 / -6.66
axle_damp_scale	0.75	wheelset modal damping 0.1-2% (Chaar 2007: measured below 2%; IPME paper 0.1-1%; snippets)	+0.57	-2.59 / +6.90	-2.59 / +6.98

Budget (RSS of worst-side 1σ terms)	Unsprung mass ±3%	Unsprung mass ±1%
Nominal fit	6.1%	5.1%
k_f, k_H0 co-estimated	3.9%	1.8%

Large-sample (pseudo-true) biases. The ±3% unsprung-mass range is an assumption with no source; ±1% corresponds to mass known from drawings.

and k_{H0} with k_{B0} reduces it to 10.0% and 7.3%. The absolute level of k_{B0} is therefore reported only with this budget; the monitoring output is the relative change.

4.6. Parameter ranges of the representative vehicle

Table 7 repeats the bound and the detection limit at the low and high end of the published (or, where none was found, assumed) range of each parameter, one at a time, with data generated and fitted by the same model. At the nominal values the support mode is at 407 Hz, the bound of k_{B0} is 0.17% and the uncalibrated two-period MDC is 0.69% ($N = 50$, 40 dB). Over all ranges except the extension of k_{B0} discussed below, the bound lies between 0.12 and 0.43%, the MDC between 0.48 and 1.77% and the support mode between 358 and 666 Hz.

Two parameters dominate. The first is the primary damping: at the laden value of c_p the MDC is 1.75%, against 0.69% at the empty value, because the strongly damped laden suspension widens the support resonance. The model uses one value of c_p for all payloads; with the laden damping acting on loaded passes only, the bound for a mixed-payload period lies between the two. The second is the support stiffness itself: a stiffer support moves the resonance towards the upper edge of the 700 Hz analysis band (666 Hz and an MDC of 1.77% at 3 GN/m), and at the bearing-only stiffness of 5 GN/m, the upper end of the extended range, the resonance reaches the band edge (702 Hz) and the MDC rises to 5.27%. A softer support (0.5 GN/m) lowers the MDC to 0.50%.

The empty axle load of 5.8–6.6 t found for Eaos/Eanos wagons, instead of the 5.5 t of the design, leaves the bound unchanged (0.17–0.17% against 0.17%). For the 1400–1526 kg wheelset masses of the multibody models, the bound is 0.18–0.19% with the additional mass on the wheels and 0.15–0.14% with it lumped on the journals; in the latter case the axlebox participation moves to a second symmetric mode near 498 Hz, while the frequency of the 407 Hz mode changes by less than 2%. The track parameters (k_f, η_f, k_{H0}) change the bound by at most a few hundredths of a percentage point.

4.7. Nonlinear Hertzian contact

Data and estimator otherwise share the linear model, so the Monte Carlo results above contain no error from the contact law itself. To test this, 20 datasets of the headline design (40 dB) were generated with nonlinear Hertzian

Table 7
Bound and detection limit over the published (or assumed) range of each parameter

Parameter	Low	High	Support mode (Hz), low / high	CRB k_B0 (%), low / high	MDC (%), low / high	Range
Nominal	–	–	407	0.17	0.69	–
Effective support stiffness k_B0 (N/m)	5e+08	3e+09	358 / 666	0.12 / 0.43	0.50 / 1.77	assumed, swept
Axlebox mass (kg)	100	158.4	407 / 367	0.17 / 0.14	0.69 / 0.57	sourced range
Primary stiffness k_p (N/m)	1.017e+06	2.28e+06	407 / 407	0.17 / 0.17	0.69 / 0.69	full text: empty / loaded (Dusza et al. 2024)
Primary damping c_p (N s/m)	7000	1.233e+05	407 / 407	0.17 / 0.42	0.69 / 1.75	full text: empty / laden (Dusza et al. 2024)
Wheel diametral inertia (kg m^2)	10	30	407 / 407	0.21 / 0.13	0.86 / 0.55	assumed, swept
Axle modal damping ratio	0.001	0.02	407 / 407	0.12 / 0.19	0.48 / 0.78	sourced range
Wheel radius (m)	0.42	0.46	407 / 407	0.17 / 0.17	0.72 / 0.69	sourced range
Support damping c_B (N s/m)	3000	1.2e+04	407 / 407	0.16 / 0.18	0.66 / 0.74	assumed, swept (factor 2 about the sourced 6740 N s/m)
Foundation stiffness k_f (N/m^2)	1.23e+08	1.333e+09	407 / 407	0.17 / 0.15	0.71 / 0.61	sourced range (derived)
Foundation loss factor	0.1	0.25	407 / 407	0.17 / 0.17	0.69 / 0.69	sourced range
Contact stiffness k_H0 (N/m)	1.388e+09	2.3e+09	407 / 407	0.17 / 0.17	0.69 / 0.69	Hertz, S1002 transverse radius +0.5 m (low); published spread up to 2.3e9 (high)
k_B0, extended to bearing-only stiffness (N/m)	5e+08	5e+09	358 / 702	0.12 / 1.27	0.50 / 5.27	extension to the EMU bearing-only vertical stiffness 4.96e9 N/m
wheelset_mass_wheels	1400	1526	400 / 399	0.18 / 0.19	0.76 / 0.77	wheelset mass 1400 kg (Dusza et al. 2024) and 1526 kg (Y25 MBS model); nominal
wheelset_mass_journals	1400	1526	498 / 495	0.15 / 0.14	0.63 / 0.60	as above, extra mass lumped on the journals
Empty axle load (t; design 5.5)	5.8	6.6	407 / 407	0.17 / 0.17	0.69 / 0.69	sourced range (empty axle load = Eanos tare 23.1-26.5 t / 4); design value 5.5 t

One parameter at a time; data generated and fitted with the same model (bound, not misspecification). 4-wheel design, N = 50, 60–110 km/h, 40 dB. MDC = (1.645 + 1.282) $\sqrt{2}$ CRB, uncalibrated. Support mode at 14 t.

contact, $F = C_H[\delta_0 + \delta]_+^{3/2}$, solved in the time domain by a fixed-point iteration with the linear vehicle–track operators, and fitted with the frozen linear estimator. The same datasets were also generated with linear contact (identical roughness, speeds, payloads and noise), so that the effect of the nonlinearity is isolated dataset by dataset. With nonlinear contact the bias of k_{B0} is 0.16 ± 0.04% and the RMSE 0.24% (linear contact: 0.10 ± 0.04% and 0.19%); the paired difference due to the nonlinearity is 0.059 ± 0.008 percentage points (at most 0.190 in any dataset), small against the detection limit. With the 40 dB simulation calibration the intervals cover the truth in 0.85 of datasets [0.62, 0.97] (linear contact 0.90). The dynamic contact deflection reaches 28.8% of the static deflection (rms, worst pass), and brief loss of contact occurs in some empty passes (at most 0.52% of the samples of a pass). For 20 pairs of periods with no change and nonlinear contact, the calibrated test raised 1 false alarms (95% interval of the rate 0.00–0.25).

Generator check. With linear contact, data and estimator share the linear model, yet the bias of k_{B0} (0.10%) and of c_B (12.8%) is larger than with the frequency-domain generator of the headline study. The two generators were therefore compared directly. Their operators agree to 0.8% (largest relative difference in the analysis band, one wheel), a noise-free record generated by both from the same roughness input differs by 0.08% (relative L_2), and the half-power damping of the support resonance is the same in both (0.055). On 10 datasets generated by both with the same speeds, payloads and roughness levels, however, the estimates from the time-domain data are higher by 0.13 ± 0.05 percentage points for k_{B0} and 10 ± 4 percentage points for c_B (frequency-domain data: -0.02% and 5.9%). The only systematic spectral difference found is a mean periodogram lower by 1.8 ± 0.7% at 480–550 Hz over 200 noise-free realisations; its cause was not identified (refining the roughness grid and the estimator's frequency table did not remove it). This dataset-level difference accounts for the whole of the linear-contact bias of k_{B0} and for most of the c_B bias. It cancels in the paired difference between nonlinear and linear contact, which uses the same generator, so the effect of the contact nonlinearity stated above is unaffected.

4.8. Payload-dependent primary suspension

The primary stiffness and damping of the Y25 bogie differ strongly between the empty and the laden wagon (Table 1), whereas the model uses one value of each. As a stress test, data were generated with both varying linearly with axle load between the empty and laden values, and fitted with fixed values.

Mixed payloads. With passes over the whole payload range and the empty values in the fit, the large-sample bias of k_{B0} is 0.70% (k_f, k_{H0} re-estimated), but individual periods are biased by -0.8 to 3.0%, depending on the payload mix of the period, and the errors do not cancel between periods: in 3 pairs of periods with no change the apparent change reached $|z| = 2.5$–7.4, and 2 pairs raised a false alarm. The standard error of the change also grew to 0.40%. Mixed-payload pooling with a fixed primary suspension is therefore not used.

Payload bands. With the comparison restricted to one payload band and the estimator using that band's fixed values (Section 3.5), the same stress truth gives a large-sample bias of 0.20% (empty band) and 0.51% (laden band). In 10 zero-change pairs in the empty band, z ranged from -1.1 to 2.1 with 0 false alarms (95% interval of the rate 0.00–0.31), and the standard error of the change was 0.26%, as without the stress; the apparent changes lean upwards (mean 0.18%, 6 of 10 positive), which the one-sided test for a loss of stiffness does not flag. In 5 laden pairs there was no false alarm either; in 2 of them the observed information was singular because c_B ran to its lower bound, and with the median standard error used for those, z ranged from -0.7 to 3.0 (largest apparent change an increase of 1.8%). The detection limits per band, from the Cramér–Rao bound with k_f and k_{H0} re-estimated, are 0.75–0.83% ($N = 50$) and 0.35–0.39%

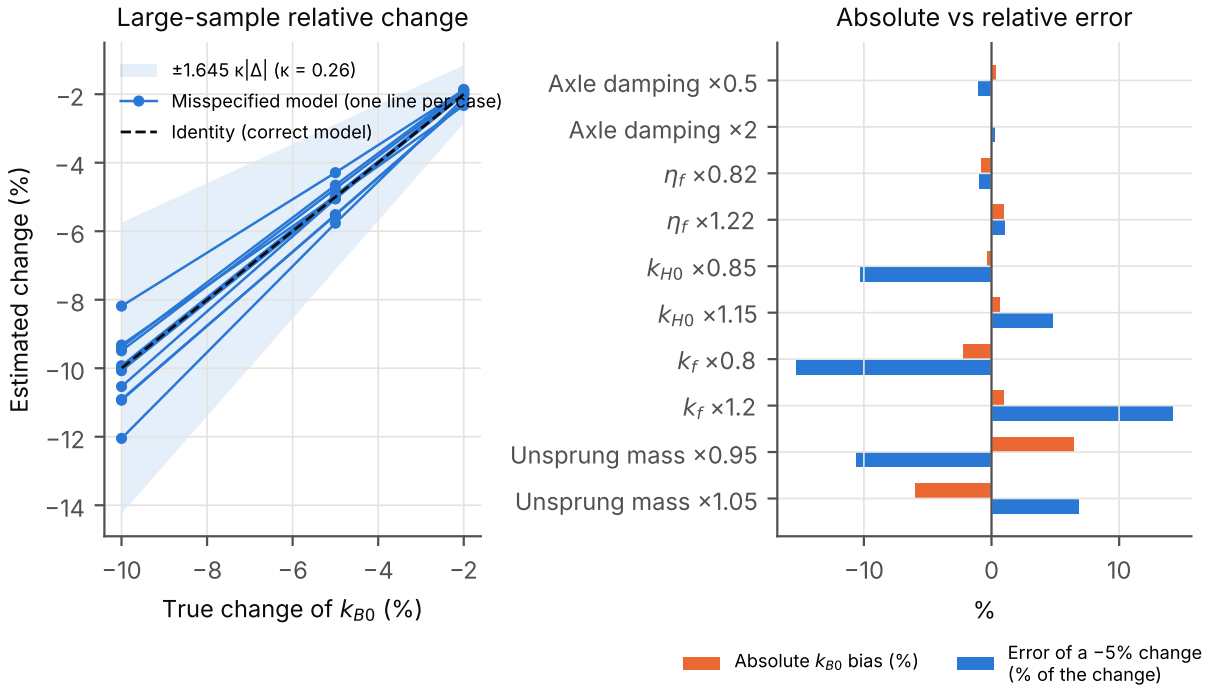


Figure 6: Relative change under model errors. Estimated against true relative change of k_{B0} under eleven single model errors (left) and the absolute bias of k_{B0} against the error of a −5% change (right). Data: `results/p2/relative_change.json`.

($N = 200$) for the empty band, and 2.04–2.14% and 0.94–0.98% for the laden band, where the strongly damped laden suspension widens the support resonance; in the laden band the support damping c_B is not resolved. Payload diversity therefore remains a lever only within a band, and the empty band is the preferred operating mode.

4.9. Relative change, interval and detection

A model error that biases $\log \hat{k}_{B0}$ by a constant cancels in the change D; what remains scales with D (Fig. 6, Table 8). With k_f and k_{H0} fixed, the gain uncertainty from one-at-a-time pseudo-true fits is $\kappa = 0.26$, dominated by the foundation stiffness (0.24). On a fresh Monte Carlo with all five model errors drawn at random it under-covers: 0.70 [0.46, 0.88], with an rms error of the change of 0.42 of Δ, because the gain grows faster than linearly in k_f. Re-estimating k_f and k_{H0} in each period lowers the one-at-a-time value to $\kappa = 0.044$; on a fresh Monte Carlo (20 replicates, k_f drawn from ×0.58 to ×2.19) the interval covered 14/20 [0.46, 0.88]. The rms error of the change was 0.096 of Δ with a mean of 0.057 (the change is under-estimated by about 6%); it does not grow with the k_f error. With the 40 dB simulation calibration applied to the standard error of D (the level bias cancels), coverage stays at 14/20 [0.46, 0.88]: the remaining misses are errors in the gain, not in the standard error.

Minimum detectable change. For a one-sided test at 5% false alarm and 90% detection, the MDC with k_f and k_{H0} re-estimated is 0.73% for $N = 50$ passes per period at 40 dB and 0.84% at 20 dB; with $N = 200$ it is 0.37% and 0.42%. With the simulation calibration (40 dB only) these become 0.78% ($N = 50$) and 0.39% ($N = 200$); 20 dB is unchanged (Table 9).

4.10. Health relevance

For the Y25 cylindrical roller bearings under the series reading (1) (initial clearance 130 µm), a 10 µm growth of the diametral clearance lowers the contact stiffness by -2.8% (empty) and -1.5% (loaded), and the effective support stiffness by -0.74% and -0.25% (Table 10). The effective change reaches the detection limit of 0.75% ($N = 50$, 40 dB, co-estimating fit) after 11 µm of growth at empty load and 37 µm at full load. Over the swept initial clearance (0–180 µm), roller count and roller length, the change per 10 µm ranges from -0.89 to -0.11% (empty) and -0.30 to -0.07%

Table 8
Relative change Δk_B0/k_B0: model-form gains, κ and interval coverage

Parameter	Gain γ per unit log error (nominal)	γ at $\Delta = -2/-5/-10\%$	κ term (1σ)	γ with k_f, k_H0 co-estimated
unsprung_scale	+1.782	+1.94, +1.79, +1.61	0.053	+1.313
kH0	+0.532	+0.58, +0.51, +0.50	0.080	–
axle_damp_scale	-0.020	-0.16, +0.01, +0.09	0.010	-0.018
kf	+0.684	+0.30, +0.75, +1.01	0.240	–
eta_f	+0.046	+0.04, +0.05, +0.04	0.009	+0.086

Quantity	Value
κ, nominal (mass ±3% / ±1%)	0.258 / 0.253
κ, k_f and k_H0 co-estimated	0.044
Combined-interval coverage, control (n=6)	1.00 [0.54, 1.00] (statistical only 0.67)
Combined-interval coverage, kf_x1.2 (n=6)	1.00 [0.54, 1.00] (statistical only 0.00)
Combined-interval coverage, unsprung_x1.05 (n=6)	1.00 [0.54, 1.00] (statistical only 0.33)
Fresh MC, misspecifications drawn from 1σ ranges (n=20): combined coverage	0.70 [0.46, 0.88] (statistical only 0.35)
Fresh MC: rms absolute k_B0 bias / rms relative-change error (fraction of Δ)	10.16% / 0.420

Estimated change error = $\gamma \cdot$ log(parameter error) $\cdot\ \Delta$; κ = RSS of $\gamma{\cdot}\sigma$; combined 90% interval = ±1.645·sqrt(SE^2 + $(\kappa\Delta)^2$).

Table 9
Minimum detectable decrease of k_B0 (5% false alarm, 90% detection)

N per period	SNR (dB)	Model-form term	SD of Δ at Δ = 0 (%)	MDC from CRB (%)	MDC with RMSE/CRB inflation (%)	Inflation
50	20	mass3pct	0.24	0.81	0.81	1.00
50	20	mass1pct	0.24	0.81	0.81	1.00
50	20	no_model_form	0.24	0.71	0.71	1.00
50	40	mass3pct	0.24	0.80	0.94	1.18
50	40	mass1pct	0.24	0.79	0.94	1.18
50	40	no_model_form	0.24	0.70	0.83	1.18
200	20	mass3pct	0.12	0.41	0.41	1.00
200	20	mass1pct	0.12	0.41	0.41	1.00
200	20	no_model_form	0.12	0.36	0.36	1.00
200	40	mass3pct	0.12	0.39	0.46	1.18
200	40	mass1pct	0.12	0.39	0.46	1.18
200	40	no_model_form	0.12	0.34	0.40	1.18

Simulation-calibrated MDC (frozen estimator; 40 dB only)

N per period	SNR (dB)	MDC (%)	SE inflation	Calibrated MDC (%)	Source
50	20.0	0.84	1.00	0.84	p4/mdc_bounds.json (coest, CRB, current config)
50	40.0	0.73	1.07	0.78	p4/mdc_bounds.json (coest, CRB, current config)
200	20.0	0.42	1.00	0.42	p4/mdc_bounds.json (coest, CRB, current config)
200	40.0	0.37	1.07	0.39	p4/mdc_bounds.json (coest, CRB, current config)

(loaded), and the growth that reaches the detection limit from 4 to 20 µm (empty) and 30 to 43 µm (loaded). The changes are not smooth in the clearance, because the number of rollers in the load zone changes in steps. For the secondary case, a double-row tapered roller unit with the geometry of [26], the corresponding growths are 6–10 and 28–39 µm. Empty passes are therefore the more sensitive to clearance growth; a mixed-payload period gives a change between the two. The other causes of support-stiffness loss (fits, housing, adapter, pad) are not modelled here; their stiffness change enters k_B undiluted.

4.11. Foundation drift

A seasonal change of the foundation stiffness between periods is the main false-alarm risk (Table 11). With k_f fixed, a later foundation 10% or 20% softer appears as a change of -2.75% or -6.05% in k_{B0} (false-alarm rate 1.00 at 10%), and a stiffer one masks real losses (MDC 2.3% and 2.0%). Re-estimating k_f and k_{H0} per period keeps the MDC at 0.75% for every drift level. A Monte Carlo with the later foundation 20% softer and no true change (20 replicates per estimator) confirms this: 20/20 false alarms with k_f fixed ([0.83, 1.00], mean apparent change -5.85%) against 0/20 with k_f re-estimated ([0.00, 0.17], mean 0.07%).

4.12. Wheel roughness at rotation orders

Wheel out-of-roundness puts lines at the rotation orders of an angle-synchronous record (Table 12). Removing the order bins (notch) costs about 2% of the bound (0.23% without wheel roughness). Modelling them as lines with a mean shared across passes lowers the bound to 0.18% when the lines are at the roughness level and 0.10% when they are 10 dB above it, because the lines then carry information about $|H|$ at known frequencies.

4.13. Repeated passes over one section

When passes repeat over the same stretch, their roughness is partially coherent. For coherence 0.5–0.9 with random lateral offsets, the independent-pass estimator stays unbiased (at most 0.09%), and its model standard error is conservative (RMSE/SE 0.64–0.82); the bin-clustered sandwich standard error is calibrated (RMSE/SE 1.08–1.10) and is used for repeated sections (Table 13).

Table 10
Health relevance: clearance growth, effective support stiffness and the load exponent

(a) Roller-contact stiffness and effective support stiffness against diametral clearance growth

Bearing	Variant	Initial clearance (μm)	Load	Bearing load (kN)	k_contact (GN/m)	Contact: change, first 10 μm (%)	Contact: mean per 10 μm, 0–50 μm (%)	Contact: growth at MDC (μm)	k_B/k_contact	Effect
cylindrical	nominal geometry	130	empty	22	3.86	-2.79	-1.87	3.0	0.26	-0.74
cylindrical	nominal geometry	130	loaded	105	5.96	-1.46	-1.02	4.4	0.17	-0.25
cylindrical	nominal geometry	0	empty	22	6.17	-5.22	-4.69	3.1	0.16	-0.88
cylindrical	nominal geometry	0	loaded	105	7.22	-0.50	-1.41	12.4	0.14	-0.07
cylindrical	nominal geometry	50	empty	22	4.72	-3.81	-2.62	0.6	0.21	-0.83
cylindrical	nominal geometry	50	loaded	105	6.71	-1.73	-1.47	3.9	0.15	-0.26
cylindrical	nominal geometry	100	empty	22	4.10	-2.52	-2.01	2.2	0.24	-0.62
cylindrical	nominal geometry	100	loaded	105	6.21	-1.40	-1.39	1.7	0.16	-0.23
cylindrical	nominal geometry	180	empty	22	3.50	-0.40	-0.91	14.8	0.29	-0.11
cylindrical	nominal geometry	180	loaded	105	5.66	-1.68	-1.14	7.1	0.18	-0.30
cylindrical	z = 15.0	130	empty	22	3.39	-2.96	-1.40	3.1	0.29	-0.89
cylindrical	z = 15.0	130	loaded	105	5.21	-1.24	-1.02	0.8	0.19	-0.24
cylindrical	z = 21.0	130	empty	22	4.23	-0.56	-1.34	11.7	0.24	-0.13
cylindrical	z = 21.0	130	loaded	105	6.67	-1.47	-1.25	7.3	0.15	-0.22
cylindrical	l = 40 mm	130	empty	22	3.45	-2.77	-1.82	7.2	0.29	-0.82
cylindrical	l = 40 mm	130	loaded	105	5.24	-0.46	-0.96	11.0	0.19	-0.09
cylindrical	l = 56 mm	130	empty	22	4.22	-2.86	-1.94	2.1	0.24	-0.69
cylindrical	l = 56 mm	130	loaded	105	6.63	-1.48	-1.11	2.5	0.15	-0.23
tapered	proxy geometry	20	empty	22	5.73	-5.68	-3.89	0.9	0.17	-1.04
tapered	proxy geometry	20	loaded	105	7.49	-1.86	-1.76	3.6	0.13	-0.25
tapered	proxy geometry	0	empty	22	6.54	-5.80	-4.96	2.8	0.15	-0.93
tapered	proxy geometry	0	loaded	105	7.65	-0.55	-1.50	11.5	0.13	-0.07
tapered	proxy geometry	50	empty	22	4.92	-3.61	-2.81	1.3	0.20	-0.76
tapered	proxy geometry	50	loaded	105	7.07	-1.72	-1.64	4.1	0.14	-0.25

(b) Effective load exponent of the series model (reading B)

Clearance growth (μm)	p_eff (least squares over payload range)	p_eff (two-point)	k_B empty / loaded (GN/m)
0.0	0.0608	0.0620	0.916 / 1.000
10.0	0.0622	0.0651	0.910 / 0.998
20.0	0.0637	0.0660	0.907 / 0.995

(c) Bound on the load exponent (k_B0, c_B, p free; 4-wheel rail-coupled, 40 dB)

p at truth	N	CRB p (abs.)	CRB k_B0, p free (%)	CRB k_B0, p fixed (%)	Two-period MDC of p	Δp 10 μm (z)	Δp 20 μm (z)	Detectable 10 / 20 μm
0.100	50	0.0031	0.25	0.17	0.0126	0.0015 (0.5)	0.0030 (1.0)	no / no
0.100	200	0.0017	0.13	0.08	0.0069	0.0015 (0.9)	0.0030 (1.8)	no / no
0.061	50	0.0030	0.25	0.17	0.0126	0.0015 (0.5)	0.0030 (1.0)	no / no
0.061	200	0.0017	0.13	0.08	0.0069	0.0015 (0.9)	0.0030 (1.8)	no / no

Main case: Y25 cylindrical roller bearings (two rows per journal); roller count and length assumed from the 130×240×80 envelope and swept; initial clearance 130 μm = lower unmounted value (operating clearance not found). Secondary case: proxy double-row tapered unit. k_housing is set so that the effective stiffness equals k_B0 at W0. MDC = 0.75% (N = 50, 40 dB, co-estimating fit). z = Δp / CRB of a single period; detection between two periods needs $\Delta p \geq 2.93 \cdot \sqrt{2} \cdot \mathrm{CRB}$.

Table 11
Seasonal foundation-stiffness drift between baseline and later period

Estimator	Later-period k_f	Apparent change, true Δ = 0 (%)	Bias at true Δ = −5% (pp)	SD of Δ (%)	False-alarm rate (nominal 5%)	MDC (%)
nominal	×0.8	-6.05	-6.17	0.24	1.000	–
nominal	×0.9	-2.75	-2.78	0.24	1.000	–
nominal	×1.1	+1.71	+1.48	0.24	0.000	2.30
nominal	×1.2	+1.67	+0.64	0.24	0.000	1.96
coest	×0.8	-0.00	+0.00	0.26	0.050	0.75
coest	×0.9	+0.00	-0.00	0.26	0.050	0.75
coest	×1.1	-0.00	-0.00	0.26	0.050	0.75
coest	×1.2	+0.00	+0.00	0.26	0.050	0.75
nominal	×1.0 (no drift)	0	0	0.24	0.050	0.70
coest	×1.0 (no drift)	0	0	0.26	0.050	0.75

N = 50 passes per period, 40 dB. 'nominal' fits (k_B0, c_B) with k_f, k_H0 at nominal values; 'coest' co-estimates k_f and k_H0 per period.

Table 12
Wheel roughness at rotation orders: notch vs tachometer nuisance lines

(a) Bound, N = 50, 40 dB

Speeds	Wheel-line level above roughness (dB)	Method	CRB k_B0 (%)	CRB c_B (%)
fixed_100	–	no_wheel_roughness	0.624	21.5
fixed_100	0	notch	0.639	22.0
fixed_100	0	tacho	0.489	16.6
fixed_100	10	notch	0.640	22.0
fixed_100	10	tacho	0.242	8.2
uniform_60_110	–	no_wheel_roughness	0.228	7.5
uniform_60_110	0	notch	0.232	7.7
uniform_60_110	0	tacho	0.179	5.2
uniform_60_110	10	notch	0.232	7.7
uniform_60_110	10	tacho	0.096	2.3

(b) Monte Carlo, 60–110 km/h

Level (dB)	Method	n	Bias (%) ± SE	RMSE/CRB	Wald 90% cov. [95% CI]
0	ignore	10	-0.01 ± 0.07	0.89	0.90 [0.55, 1.00]
0	notch	10	0.02 ± 0.09	1.21	0.80 [0.44, 0.97]
0	tacho	10	-0.02 ± 0.06	0.99	0.80 [0.44, 0.97]
10	ignore	10	-0.11 ± 0.06	0.96	0.90 [0.55, 1.00]
10	notch	10	0.06 ± 0.07	0.87	0.90 [0.55, 1.00]
10	tacho	10	0.02 ± 0.04	1.19	0.80 [0.44, 0.97]

4.14. Validation against the P2 failure mode

Carrigan and Talbot [3] estimate the foundation stiffness from the P2 peak of single passes and call for a reliable way to reject bad estimates. On their simulated setup with a resilient foundation (Fig. 7, Table 14), corrugation near the P2 frequency puts 46% of per-pass estimates outside ×0.7–×1.43 of the reference (median 9.36 MN/m^2; 11% without corrugation). The multi-pass likelihood, fitting k_f instead of k_{B0}, gives a median of 7.74 MN/m^2 with corrugation and 7.71 MN/m^2 without (truth 7.7), but its profile interval covers the truth in only 0.45 of sections with corrugation (0.80 without).

Table 13
Repeated passes over one section with partial coherence (k_B0)

(a) Independent-pass estimator on partially coherent data

ρ	n	Bias (%) ± SE	RMSE (%)	Variance	Median SE (%)	RMSE/SE	90% cov. [95% CI]
0.5	12	0.01 ± 0.06	0.19	model SE	0.23	0.82	0.83 [0.52, 0.98]
0.5	12	0.01 ± 0.06	0.19	sandwich, bin clusters	0.17	1.10	0.83 [0.52, 0.98]
0.5	12	0.01 ± 0.06	0.19	sandwich, pass clusters	0.15	1.24	0.83 [0.52, 0.98]
0.7	12	0.04 ± 0.05	0.19	model SE	0.23	0.82	1.00 [0.74, 1.00]
0.7	12	0.04 ± 0.05	0.19	sandwich, bin clusters	0.17	1.09	0.75 [0.43, 0.95]
0.7	12	0.04 ± 0.05	0.19	sandwich, pass clusters	0.13	1.41	0.83 [0.52, 0.98]
0.9	12	0.09 ± 0.03	0.15	model SE	0.23	0.64	1.00 [0.74, 1.00]
0.9	12	0.09 ± 0.03	0.15	sandwich, bin clusters	0.14	1.08	0.92 [0.62, 1.00]
0.9	12	0.09 ± 0.03	0.15	sandwich, pass clusters	0.13	1.12	0.83 [0.52, 0.98]

(b) Bounds and the exchangeable-ρ likelihood

ρ	Oracle CRB (%)	Exchangeable, ρ estimated: CRB (%)	Independent CRB (%)	Exchangeable MC bias (%) ± SE	Exchangeable MC RMSE (%)	Exchangeable Wald cov.
0.5	0.005	0.182	0.225	0.08 ± 0.11	0.37	0.67 [0.35, 0.90]
0.7	0.004	0.156	0.225	0.09 ± 0.06	0.21	0.75 [0.43, 0.95]
0.9	0.005	0.120	0.241	0.10 ± 0.05	0.19	0.50 [0.21, 0.79]

The oracle assumes the coherence matrix is known exactly; it is not achievable in practice.

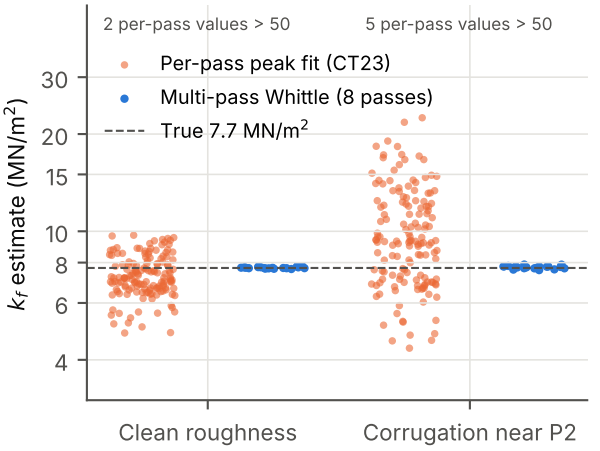


Figure 7: Track-side check against per-pass P2 fits. Foundation stiffness from per-pass P2 peak fits [3] and from the multi-pass likelihood (8 passes per section), with and without corrugation near P2. Data: `results/p2/scope_p2_hysteretic.json`.

Table 14
P2 track-stiffness validation against the CT23 per-pass failure mode (true k_f = 7.7 MN/m²)

Roughness	Per-pass median (MN/m²)	Per-pass outside x0.7–x1.43	Per-pass range	Per-pass no peak	Multi-pass median	Multi-pass range	Multi-pass profile 90% covers truth
Clean roughness	7.18	11%	4.8–605	6%	7.71	7.64–7.79	0.80
Corrugation near P2	9.36	46%	4.3–613	6%	7.74	7.58–7.89	0.45

With corrugation the multi-pass point estimate stays within 2% of the truth, but its profile interval under-covers (0.45 of 90%); the cause is not yet diagnosed (open item).

5. Discussion

What the method delivers. The estimator separates the support resonance from the roughness because the two scale differently with speed: the resonance is fixed in frequency, the roughness excitation is fixed in wavenumber. Speed spread is therefore the design variable that matters most (Section 4.2); at a single speed the support stiffness is barely identifiable, and a spread of 60–110 km/h, typical of a mixed freight route, brings the bound to a fraction of a per cent with 50 passes. Payload variation and a weigh-in-motion measurement add the load dependence of the contact and support stiffness. Neither requires a dedicated test: the passes are those a wagon makes in service, and the roughness never has to be measured.

Absolute level and relative change. The absolute level of k_{B0} inherits the error of every other model parameter, above all the unsprung mass and the foundation stiffness (Section 4.5); its budget of a few per cent is an order of magnitude larger than the statistical error. For monitoring this matters less than it seems, because a model error that is the same in two periods largely cancels in the change. What does not cancel is a change of the track between periods,

Table 15
Validation: κ revalidation, drift at k_f ×0.8, bias diagnosis, robust standard errors, calibration

(a) κ revalidation, co-estimating fit

Quantity	Value
n (stratified k_f draws)	20
k_f range drawn	×0.58 – ×2.19
Coverage, $\kappa = 0.044$	0.70 [0.46, 0.88]
Coverage, statistical only	0.65 [0.41, 0.85]
rms change error (fraction of Δ)	0.096
mean change error (fraction of Δ)	+0.057
rms absolute k_B0 bias, base period (%)	3.25

Worst cases

Rep	k_f	η_f	k_H0	Unsprung	Axle damping	True change (%)	Estimated (%)	Error (fraction of Δ)	Covered
1	×1.22	×0.85	×0.83	×0.99	×1.53	-4.14	-3.42	+0.177	no
5	×0.93	×1.60	×0.94	×1.04	×2.25	-9.22	-7.82	+0.158	no
13	×0.62	×0.87	×1.11	×1.03	×0.96	-2.31	-2.01	+0.133	yes
4	×0.65	×0.87	×1.20	×1.03	×1.36	-6.06	-5.32	+0.126	no
3	×0.86	×1.02	×0.96	×0.99	×1.37	-6.59	-5.83	+0.119	no

(b) Drift Monte Carlo (true change 0)

Later k_f	Estimator	n	False alarms	Rate [95% CI]	Mean Δ (%)	SD Δ (%)	Median SE (%)
×0.8	nominal	20	20	1.00 [0.83, 1.00]	-2.21	0.35	0.27
×0.8	coest	20	1	0.05 [0.00, 0.25]	-0.02	0.22	0.24
×0.8	nominal	20	20	1.00 [0.83, 1.00]	-5.85	0.81	0.25
×0.8	coest	20	0	0.00 [0.00, 0.17]	+0.07	0.25	0.25

(c) Bias diagnosis on paired datasets (same data, different SNR or estimator)

Seed	Estimator	SNR (dB)	n	Bias (%) ± SE	RMSE (%)	CRB (%)	RMSE/CRB	Wald cov.
42	base	20	10	-0.068 ± 0.080	0.249	0.179	1.40	0.90
42	base	40	10	+0.021 ± 0.075	0.226	0.170	1.33	0.90
42	base	60	10	+0.022 ± 0.075	0.227	0.170	1.33	0.80
42	refined	40	10	+0.021 ± 0.075	0.225	0.170	1.32	0.80
123	base	40	10	+0.277 ± 0.047	0.310	0.169	1.83	0.40
123	base	60	10	+0.284 ± 0.047	0.316	0.169	1.87	0.30
123	kb120	40	10	+0.219 ± 0.046	0.259	0.169	1.53	0.70
123	kb120_K1	40	10	+0.046 ± 0.284	0.854	0.164	5.22	0.20
123	kb120_K4	40	10	+0.239 ± 0.059	0.297	0.169	1.76	0.70
123	refined	40	10	+0.274 ± 0.047	0.307	0.169	1.81	0.40

(d) Regression of the 50 headline errors (40 dB) on pass-design features

Feature	Coef. (% per SD)	t	p	r alone (p)
speed_sd_kmh	-0.044	-1.39	0.171	-0.23 (0.110)
payload_sd_t	+0.064	+2.09	0.042	+0.33 (0.021)
speed_min_kmh	+0.031	+1.00	0.324	+0.17 (0.228)
bins_near_resonance	+0.031	+1.01	0.320	+0.17 (0.247)
Intercept (mean error)	+0.146			
R^2 / F-test p	0.19		0.049	
Seed one-way ANOVA p			0.036	

(e) Wald 90% coverage of k_B0 on the 50 headline fits per SNR, by standard error

SE	20 dB: coverage [95% CI]	40 dB: coverage [95% CI]	20 dB: median SE (%)	40 dB: median SE (%)	20 dB: RMSE / median SE	40 dB: RMSE / median SE
observed information (stored)	0.88 [0.76, 0.95]	0.70 [0.55, 0.82]	0.178	0.170	1.12	1.55
expected information	0.88 [0.76, 0.95]	0.68 [0.53, 0.80]	0.177	0.170	1.12	1.55
sandwich, bin clusters	0.88 [0.76, 0.95]	0.62 [0.47, 0.75]	0.172	0.165	1.16	1.60
sandwich, 8-bin blocks	0.88 [0.76, 0.95]	0.72 [0.58, 0.84]	0.178	0.177	1.12	1.49
sandwich, 61-bin blocks	0.90 [0.78, 0.97]	0.74 [0.60, 0.85]	0.181	0.179	1.10	1.47

(f) κ replicates (n = 20): coverage of the change, $\kappa = 0.044$, slope correction 0.99998

SE of the change	Median SE (%)	Statistical only	With κ	With κ and slope correction
stored (fit)	0.250	13/20	14/20 [0.46, 0.88]	14/20
expected information	0.254	13/20	14/20 [0.46, 0.88]	14/20
sandwich, bin clusters	0.261	13/20	14/20 [0.46, 0.88]	14/20
sandwich, 8-bin blocks	0.288	14/20	14/20 [0.46, 0.88]	14/20
sandwich, 61-bin blocks	0.348	14/20	16/20 [0.56, 0.94]	16/20

(g) Calibration of the change, correct model (pseudo-true, k_f and k_H0 co-estimated)

True change (%)	Estimated change (%)
-2.0	-2.0000
-5.0	-5.0000
-10.0	-9.9998
Slope	0.99998

The slope correction and the sandwich SEs were not tuned on the κ replicates. kb120 = debiasing kernel half-width 120 bins (default 30).

and re-estimating the foundation and contact stiffness in each period removes the resulting false alarms (Section 4.11). The method is therefore best used as a trend monitor for each wheelset, with the first period as its own baseline.

Which faults can be seen. The detection limit of the change is below one per cent for 50 passes per period and roughly halves with 200 (Section 4.9). A loss of housing, adapter or pad stiffness enters k_B directly and is detectable once it exceeds this limit. Clearance growth is a weaker signal, because the roller contacts are much stiffer than the rest of the support and their change is diluted in the series sum (Section 4.10). The method is not a substitute for envelope or temperature monitoring of rolling-contact faults; it covers a different failure path, the progressive loss of support, which those methods do not see.

Relation to track monitoring. The same likelihood, with the foundation stiffness as the target, gives median estimates close to the truth where per-pass P2 fits scatter widely under corrugation (Section 4.14); its intervals then under-cover, so on the track side it can screen per-pass estimates but does not yet give calibrated intervals.

5.1. Limitations

Residual bias and simulation-calibrated intervals. For the representative wagon the frozen estimator has a bias of at most 0.07% (Table 3) and a scatter up to 1.08 times the Cramér–Rao bound, but the 40 dB Wald intervals still

cover the truth slightly less often than nominal (0.84). In the earlier configuration the bias reached 0.219% for one seed, and Section 4.4 ruled out the obvious causes: the large-sample (pseudo-true) bias is nil (slope 0.99998), sensor noise does not change the error, exact sub-band scoring and a finer table change it by at most 0.01 percentage points, robust (sandwich) standard errors do not restore coverage, and fewer roughness levels do not remove it (no incidental-parameter effect). Truncation of the debiasing kernel explained about a fifth and is removed by the wider kernel; the remainder is unexplained and depends on the realisation of roughness, speeds and payloads. The 40 dB intervals are therefore simulation-calibrated for the stated design (speed range, payload range, number of passes, roughness model and noise level), and must be re-calibrated for any other design. The calibration was checked on held-out seeds (Section 4.1); at 20 dB, where no calibration is applied, the held-out coverage differed between the two splits, so the nominal 90% should be read as approximate. For relative changes under random model errors, the calibrated interval still under-covers (14/20): the gain uncertainty κ derived from one-at-a-time model errors is too small, and an interval for the change that holds under realistic model errors remains open.

Load exponent as a clearance indicator. Clearance growth changes not only the level of the effective support stiffness but also its dependence on axle load, because a larger clearance shrinks the load zone more at light load. In principle the load exponent p of (2) could therefore separate clearance growth from other losses of support stiffness. With the exponent free alongside (k_{B0}, c_B) and payloads spanning 5.5–22.5 t, the Cramér–Rao bound of p is 0.0031 for $N = 50$ passes and 0.0017 for $N = 200$, and freeing p raises the bound of k_{B0} from 0.17% to 0.25% ($N = 50$). Under the series reading of the support (Y25 cylindrical roller bearings, Table 1), 10 and 20 µm of clearance growth raise the effective exponent by only 0.0015 and 0.0030. Detecting a shift between two periods (one-sided 5% false alarm, 90% detection) needs 0.0126 at $N = 50$ and 0.0069 at $N = 200$, so neither would be detected, even with 200 passes per period. The exponent is therefore held fixed in the estimator, and the method reports a loss of effective support stiffness without attributing it to a mechanism.

Linear contact and model form. Data and estimator share the linear vehicle–track model, so the Monte Carlo results do not include the error of the model itself beyond the parameter errors of Section 4.5. The wheel–rail contact is linearised about the static load; the effect of the nonlinear Hertzian contact on the estimates is reported in Section 4.7. That check also exposed an unexplained difference of 0.13 percentage points in k_{B0} and about 10 in c_B between two generators that implement the same linear model; it is small against the detection limit and cancels in changes measured with one generator, but it shows that the residual bias of c_B depends on details of the simulation that were not resolved. The model is vertical and two-dimensional, the bogie frame is held fixed, the wheels are rigid, and the rail pinned–pinned resonance of discretely supported track is not represented; all of these lie outside or at the edge of the analysis band but have not been tested against data.

Primary suspension and payload bands. The primary damping of the Y25 bogie comes from a friction damper; the values used are low-frequency equivalent viscous coefficients for the empty and the laden wagon [14], not validated at the 400 Hz of the support resonance, where the damper is likely to stick and act more as a stiffness than as a damper. The stress test of Section 4.8, which lets the low-frequency values vary with payload, is therefore likely to be conservative. It shows that pooling passes over the whole payload range requires a payload-dependent model of the primary suspension: with a fixed suspension, the error of each period depends on its payload mix and does not cancel in the change. The method is therefore operated within payload bands, with the empty band preferred. Co-estimating a payload-dependent primary suspension, which would restore the full payload range, is left for future work, as is measuring the damper's behaviour at the support frequency, which the absolute level of k_{B0} also depends on.

Representative parameter values. The vehicle and track values are those of a representative freight wagon (Y25 bogies, 22.5 t axle load) on ballasted 60E1 track, taken from published sources where they could be found (Table 1). The effective support stiffness, the support damping, the wheel diametral inertia and the internal geometry of the bearing were not found for a freight axlebox and are swept over plausible ranges (Table 7); the bound and detection limit are reported over each range rather than at a single value. A specific vehicle requires its own values, above all the unsprung mass.

Roughness and operating model. Roughness is modelled as a stationary Gaussian process with a level per sub-band that differs between passes but is common to the wheels of one pass, and speed is constant within a pass. Non-stationary features (joints, welds, switches) and speed changes within a record are not modelled and would have to be excluded or

windowed out in practice. Environmental effects on the support, for example the temperature dependence of elastomeric pads, would appear as changes of k_B and must be separated by comparing like conditions.

No field data. All results are from simulation. A field test with an axlebox whose support stiffness is changed in a controlled way is needed before the detection limits can be claimed for a real vehicle.

6. Conclusions

The vertical support stiffness of a wheelset bearing can be identified from in-service axle-box acceleration without measuring the rail roughness, by pooling passes at different speeds and payloads in a debiased Whittle likelihood in which the roughness levels are free nuisance parameters. For a representative freight wagon on ballasted track, simulations show that:

- with 50 passes at 60–110 km/h the support stiffness is estimated with a bias of at most 0.07% and a root-mean-square error of 0.19–0.20%, within 1.08 times the Cramér–Rao bound; intervals at 40 dB are calibrated by simulation;
- the speed spread is the design variable that makes the support stiffness identifiable, and the wheel-to-wheel coupling through the rail must be modelled, since ignoring it biases the estimate by -5.6%; nonlinear Hertzian contact, by contrast, shifts the estimate by only 0.059 percentage points;
- the absolute level carries a model-form uncertainty of about 10.0%, so the method is a trend monitor: comparing periods within a payload band and re-estimating the foundation and contact stiffness in each, a decrease of 0.75–0.83% is detectable with 50 empty passes per period, and a change of the track foundation between periods causes no bias;
- pooling passes over the whole payload range requires a payload-dependent model of the primary suspension: with a fixed suspension the error of each period depends on its payload mix and raised false alarms, while within a band it did not;
- a loss of housing, adapter or pad stiffness reaches the identified stiffness directly, while clearance growth is diluted by the stiff roller contacts, so the method reports support-stiffness loss without attributing it to a mechanism.

The next steps are a field test with a controlled change of support stiffness; co-estimation of a payload-dependent primary suspension, to restore the full payload range; application to freight wagons of Indian Railways, once vehicle-specific values are available from RDSO; an extension of the model to the pinned–pinned frequency band; and calibrated intervals for the track-side use of the same likelihood.

CRediT authorship contribution statement

[Given name 1] [Family name 1]: [Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization]. **[Given name 2] [Family name 2]:** [roles].

Declaration of competing interest

[The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.]

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Data availability

The code that generates every result, table, figure and number in this paper, and the stored result files, are available at https://github.com/Shreyansh369/wheelset-s2pinn and archived at Zenodo (https://doi.org/[ZENODODOI]). All data are simulated; no measured data were used. The repository references the third-party S^2-PINN code of Li and Onwunta [27] as a pinned external dependency; that code is not redistributed.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Claude Code (Anthropic) in order to develop the simulation and estimation code, assist the literature search, and draft and edit the manuscript text. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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